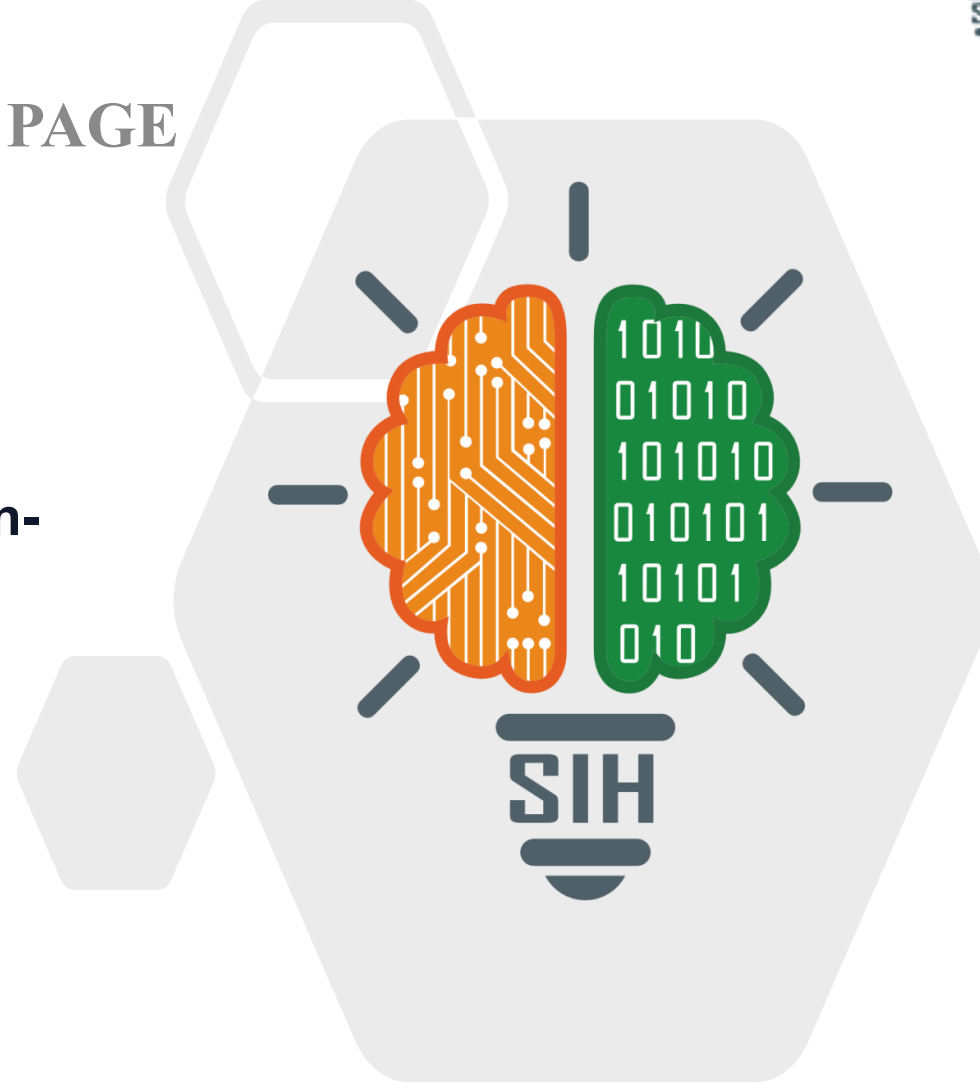


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement ID – **SIH26117**
- Problem Statement Title – **Sovereign On-Premise Agentic AI Workbench**
- Theme – **Smart Automation**
- PS Category – **Software**
- Team ID – **087**
- Team Name – **Den of Devs**



Proposed Solution

CogniShift is an air-gapped sovereign agentic AI workbench engineered for high-consequence refinery operations. It empowers plant engineers to safely parse complex P&IDs, audit inspection envelopes, and execute ASME Section VIII calculations with zero cloud egress.

Novelty

Think of it this way: Public Cloud AI Vulnerability vs. Sovereign Local Intelligence

The "Cloud AI Risk" (Public Vulnerability): Sending proprietary refinery P&IDs, SCADA registers, and operating envelopes to external cloud APIs can conflict with industrial cybersecurity, confidentiality, and data-governance requirements. WAN disconnects paralyze plant emergency response.

The "CogniShift Solution" (100% On-Site Sovereignty): An air-gapped agentic AI workbench running entirely on plant-floor edge hardware. Every inference, vector embedding, and supervisor permit executes locally with verifiable 0.00 KB/s outbound egress.

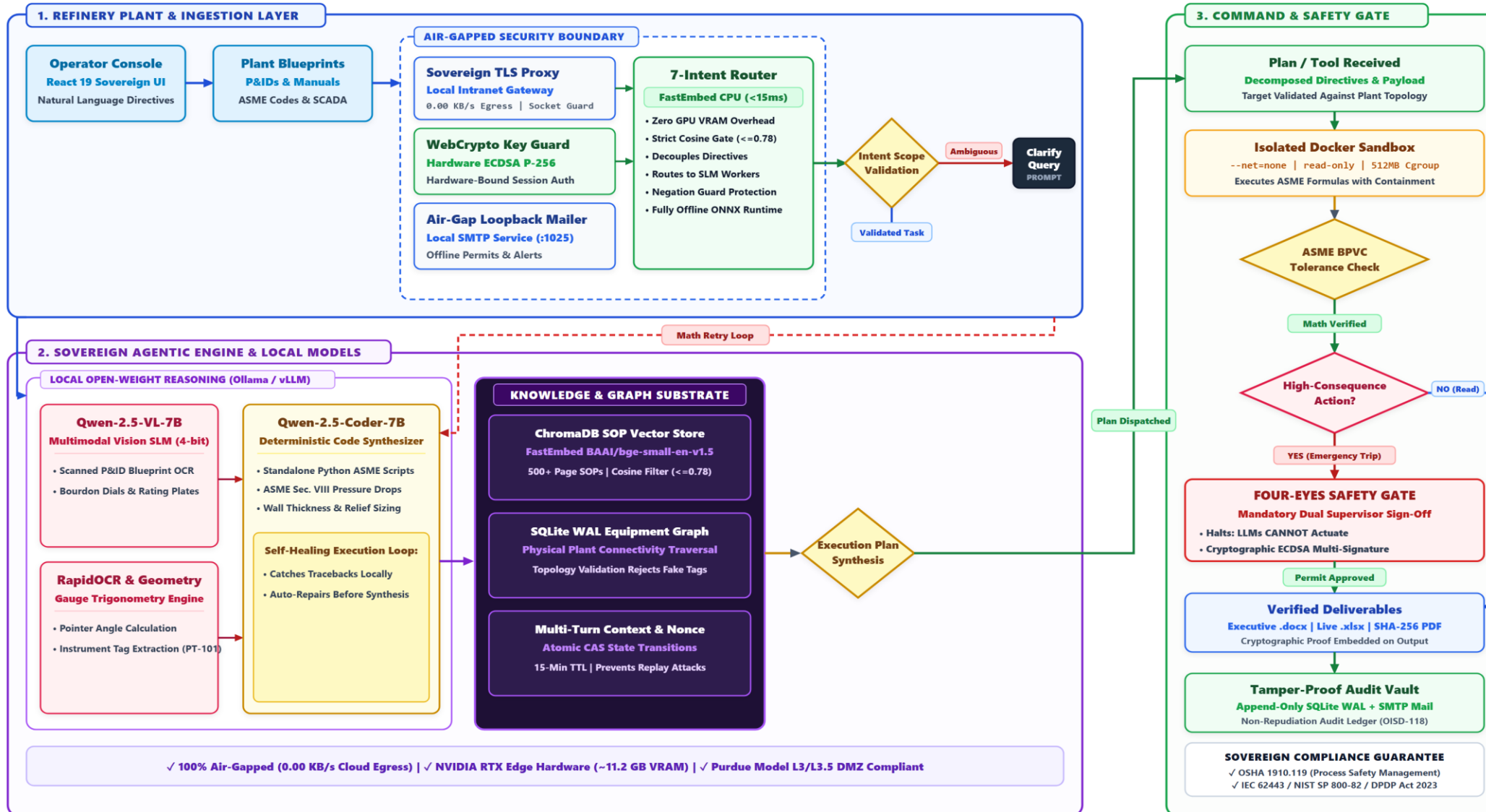
At its core, our solution provides four key abilities:

- 1. Multimodal P&ID & Dial Vision:** Parses complex refinery piping blueprints, pipe diameters, Bourdon dial angles, and instrument tags using local 4-bit vision SLMs served via Ollama / vLLM and classical geometry.
- 2. Isolated Sandboxed Math Execution:** Executes and verifies engineering calculations deterministically inside an air-gapped Docker container (--net=none) with ASME BPVC tolerance verification loops.
- 3. Hybrid Knowledge & Topology Substrate:** Combines dense semantic vector search over 500+ page SOP manuals (ChromaDB) with a physical refinery piping graph (SQLite WAL).
- 4. Deterministic Four-Eyes Safety Gate:** High-consequence operations (emergency relief, pump trips) halt execution until dual shift supervisors cryptographically sign off.



Key Features Under the Hood:

- 1. Zero Cloud Egress Guarantee (0.00 KB/s):** Continuous kernel-level socket telemetry (psutil & iptables) provides live UI readout verifying that zero bytes leave the refinery perimeter.
- 2. FastEmbed 7-Intent Semantic Router:** Decouples user directives on CPU ONNX in <15ms before engaging heavy models, preserving edge GPU VRAM and preventing model thrashing.
- 3. Hardware-Bound WebCrypto & Air-Gap Mailer:** Browser-generated ECDSA P-256 keys bind terminal sessions to authorized hardware; an internal loopback SMTP mailer (:1025) dispatches shift handovers and permits 100% offline.
- 4. Verifiable Multi-Format Deliverables:** Directly synthesizes formal Word memos, styled Excel sheets with live formulas, and publication-grade PDF audits hashed with SHA-256.



Core AI Models

- Qwen-2.5 (VL & Coder)
- Ollama / vLLM Serving
- FastEmbed BGE-Small

Execution Engine

- Isolated Docker Sandbox
- RapidOCR & Dial Geometry
- SQLite Physical Topology

Sovereign Substrate

- 100% Air-Gapped Edge Node
- ChromaDB SOP Store
- Loopback SMTP Mailer
- WebCrypto ECDSA Session
- 0.00 KB/s Outbound Egress

Operator Control

- Vite + React 19 Client
- Dual-Supervisor HITL Gate





Can We Actually Build It? (Feasibility):

Smart AI on edge hardware? Yes. We run 4-bit quantized open-weight SLMs (Qwen-2.5-VL and Qwen-2.5-Coder) requiring ~11.2 GB active VRAM, which fits within a 12 GB-class GPU under the configured quantization/resource limits.

Scalability & multi-user concurrency: Continuous-batched inference via Ollama / vLLM enables concurrent users on a single GPU, with horizontal inference-worker scale-out for larger deployments. FastAPI's async core and SQLite WAL mode support simultaneous operator laptops across the refinery LAN.

Bandwidth & cloud costs? Zero. The entire stack operates 100% air-gapped within the plant boundary. Zero recurring cloud API subscriptions, zero token egress fees, and zero dependence on external connectivity.



Will It Work in the Real World? (Viability):

Battle-tested open source stack: Built on proven primitives (Ollama / vLLM, Docker, FastEmbed, ChromaDB, React 19). Zero vendor lock-in; open-weight models can be hot-swapped as open weights advance without rewriting application logic.

Deterministic plant safety interlocks: LLMs never actuate refinery equipment directly. Dangerous commands (emergency relief, pump reboots) halt execution for dual shift supervisor cryptographic sign-off (Four-Eyes principle).

Self-healing execution loops: When sandboxed Python scripts fail ASME tolerance checks, the engine automatically catches tracebacks and repairs the script before report synthesis.



What's It For? (Operational Use Cases):

The "Engineering Co-Pilot": Ingests 500+ page ASME inspection manuals, scanned P&IDs, and SCADA stream registers. Accelerates pipe wall corrosion evaluations and relief valve sizing for rapid engineering turnarounds.

The "Sovereign Decision Hub": Real-time control room terminal for shift supervisors to inspect equipment topologies, authorize cryptographic work permits, and monitor refinery compliance ledgers.



Why Does This Matter? (Operational Value)

Eliminates cognitive fatigue: No engineer can cross-reference thousands of P&IDs and 500-page ASME standards under emergency stress. CogniShift verifies safety envelopes in seconds, freeing engineers for critical decisions.

Guarantees data sovereignty: Proprietary refinery schematics, SCADA registers, and yield formulas never leave the site perimeter. Supports DPDP-aligned data sovereignty through on-premise processing.

Unyielding industrial resilience: Operates seamlessly during WAN fiber cuts, severe cyclones, or cyber embargoes with zero external cloud API dependencies.

Potential Risks & Our Solutions

What if calculation scripts produce incorrect values?

Python scripts execute in an isolated Docker container with ASME tolerance verification loops, validating results against defined tolerance checks.

What about proprietary data leakage?

100% air-gapped on edge hardware. Verified 0.00 KB/s egress. Zero external API calls.

What if an operator sends unsafe commands?

Deterministic Four-Eyes principle. High-risk actions halt until two independent supervisors authorize.

What if edge hardware has limited VRAM?

FastEmbed CPU router (<15ms) decouples tokens; 4-bit SLMs served via Ollama / vLLM fit within a 12 GB-class GPU under configured quantization limits.

Accelerated Engineering Turnaround.

- Ingests 500+ page ASME manuals, scanned P&IDs, and SCADA logs in seconds. Slashes report preparation and remaining life calculations from hours to minutes.

Deterministic Math

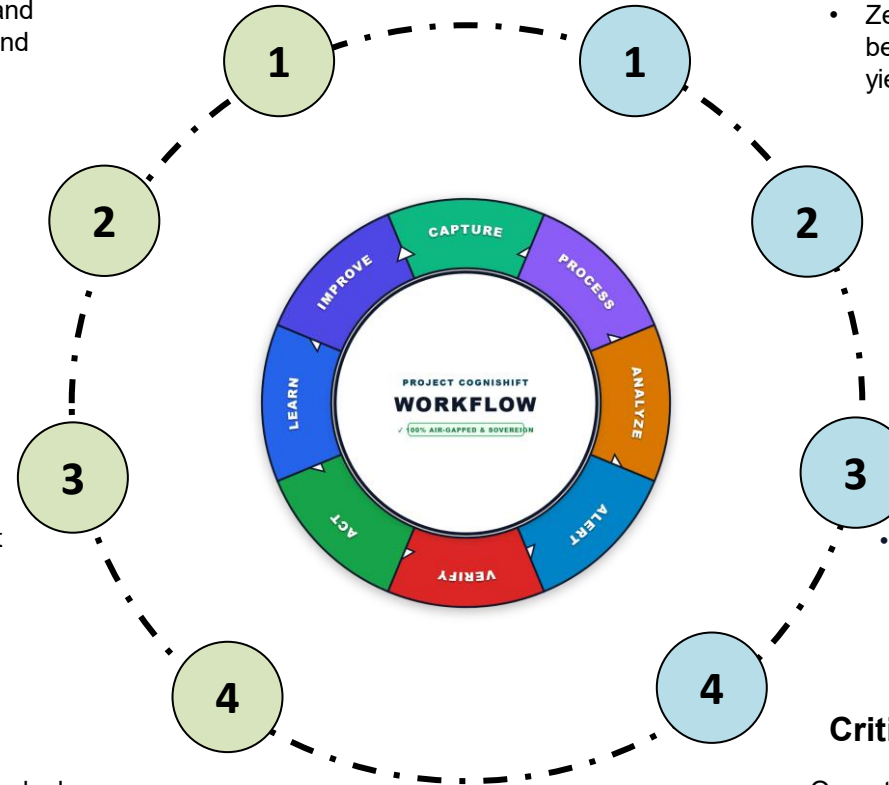
- Automated execution of pressure drops and relief margins inside an air-gapped Docker container executes calculations deterministically and validates them against defined engineering constraints.

Dual-Supervisor Safety

- Dual independent supervisor sign-offs eliminate single-point operator error. High-consequence actions cannot execute without cryptographic dual authorization.

Verified Deliverables

- Directly outputs executive .docx approval memos, styled .xlsx calculation sheets with live formulas, and publication-grade PDF audits with SHA-256 integrity hashes.



Zero Cloud Data Egress

- Zero corporate, employee, or plant telemetry data ever egresses beyond the refinery perimeter. Proprietary P&ID topologies and yield formulas remain 100% sovereign.

Massive Cost Savings

- Acts as a 24/7 force multiplier for plant maintenance teams. Slashes data egress costs to zero and eliminates recurring cloud token licensing expenses.

DPDP-Aligned Data Sovereignty.

- Supports DPDP-aligned data governance by keeping sensitive processing and storage on-premise.

Critical Plant Defense

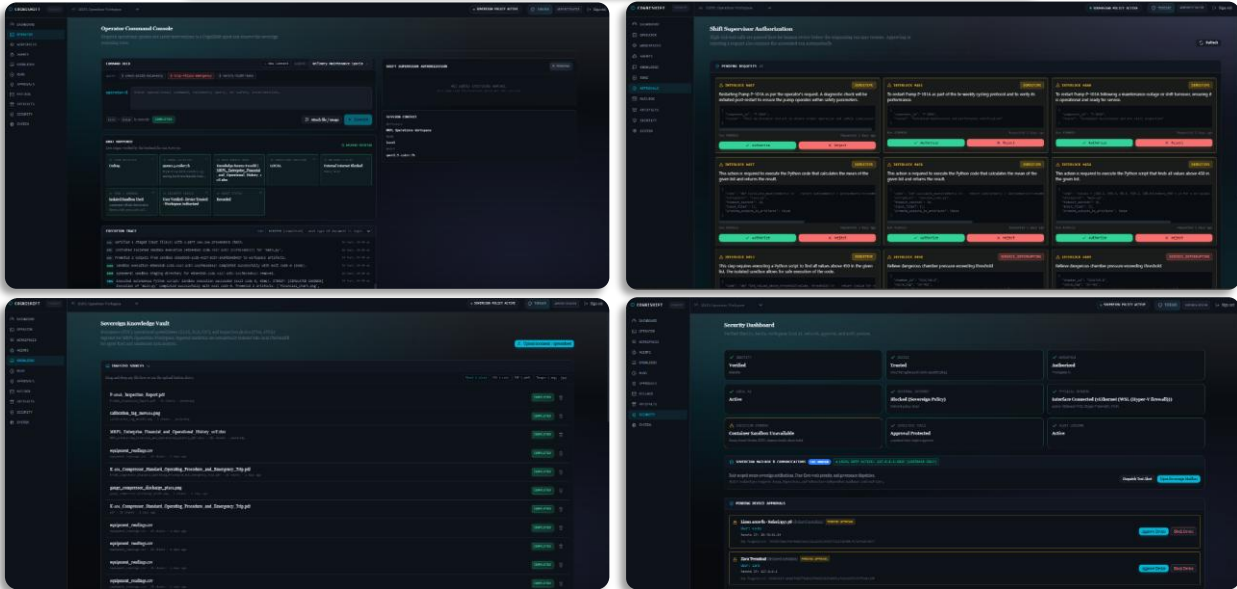
- Operates continuously during severe cyclones, fiber cuts, or geopolitical cyber embargoes. Independent of external cloud API availability during local operation.

Key Academic & Statutory References

Our sovereign architecture is grounded in statutory industrial engineering codes and peer-reviewed safety frameworks:

- [1] ASME Boiler & Pressure Vessel Code (BPVC) Sec. VIII, Div. 1 & 2 (ASME, 2023)
 - Enforces deterministic wall thickness sizing, joint efficiency limits, and pressure relief in Docker sandbox.
- [2] IEC 62443 / NIST SP 800-82: Industrial Control Systems Cybersecurity (ISA/IEC, 2021)
 - Establishes sovereign Purdue Level 3/3.5 DMZ isolation, zero cloud egress verification, and socket security.
- [3] RAGAS: Automated Evaluation of Retrieval Augmented Generation (Es et al., 2024)
 - Uses calibrated retrieval thresholds to improve evidence grounding and reduce unsupported responses.
- [4] OSHA 1910.119 (PSM) & DPDP Act 2023 / W3C WebCrypto Device Trust (IEEE/W3C, 2023)
 - Informs process-safety, data-governance, device-trust, and human-authorization controls used in CogniShift.

Sovereign Operator Workbench & Inspection UI



To review our complete project abstract, technical methodology, full system architecture, and statutory references, please scan the QR code.



Enterprise Architectural Trade-Off Analysis

We benchmarked latency, data security, and operational resilience across architectural paradigms. CogniShift combines verifiable local data sovereignty with deterministic, low-latency safety controls for governed industrial AI workflows.

Feature	CogniShift (Sovereign)	Cloud AI Solutions	Standard SCADA / COTS*
Zero Cloud Egress	Guaranteed (0.00 KB/s) ✓	Failed (Mandatory WAN) ✗	Guaranteed (No AI Model) ✗
Math & Code Safety	Certified (Docker Sandbox) ✓	Unverified (Hallucinations) ✗	N/A (Static Logic Only) ✗
Statutory Compliance	ASME-Grounded / DPDP-Aligned ✓	External Data Processing Required. ✗	Manual Engineer Audit ✗
Operating Cost	Zero Egress Fees (Local) ✓	High (Per-Token Pricing) ✗	High (Proprietary Licensing) ✗

*COTS = Commercial Off-The-Shelf | DPDP = Digital Personal Data Protection Act